

## WEEK – 14

**DURATION: 13 April 2026 – 17 April 2026**

**DAY 2: 14 April 2026**

### 1. TASKS WORKED ON

- ***Revision of React Fundamentals:*** Revised core React fundamentals, including Virtual DOM, JSX, components, props, fragments, and state management concepts. Strengthened my understanding of React rendering and component architecture through theoretical study and practical exercises.
- ***Revision of React Hooks:*** Revised key React Hooks, including `useState()`, `useEffect()`, and `useRef()`, to strengthen my understanding of state management, side effects, and DOM interactions through practical implementation.
- ***Revision of Custom Hooks:*** Studied and implemented reusable custom hooks to improve code reusability and component logic management. Revised practical examples of `useCounter()`, `useDebounce()`, `useFetch()`, and `useLocalStorage()`.
- ***Revision of Higher-Order Components (HOCs):*** Revised the concept of Higher-Order Components (HOCs) and implemented practical examples to understand component enhancement and reusable logic patterns. Explored HOCs such as `withLogger`, `withToggle`, and `withLoading`.
- ***Revision of Controlled and Uncontrolled Components:*** Revised controlled and uncontrolled component patterns in React and implemented both approaches to understand state-based and ref-based form input handling techniques.

## **2. LEARNING AND KNOWLEDGE GAINED**

Through these revision tasks, I strengthened my understanding of React fundamentals including components, hooks, custom hooks, HOCs, and form handling techniques. The practical implementation of these concepts improved my ability to build reusable, scalable, and structured React applications efficiently.

## **3. PLAN FOR NEXT WORKING DAY**

- Revise remaining advanced React concepts.
- Continue improving practical implementation skills through project-based exercises.

## **DAY 3: 15 April 2026**

### **1. TASKS WORKED ON**

- **Aeko Website – Catalogue Page Development:** Developed a Catalogue page with dynamic backend integration for displaying and managing catalogue items. Also explored file upload workflows and gained an understanding of secure upload approaches, including Presigned URLs and server-side uploads.
- **Implementation of Two-Factor Authentication (2FA):** Learned the fundamentals of OTP-based authentication and implemented a basic 2FA workflow using the MERN stack. Also gained practical understanding of backend development concepts such as API creation, request handling, validation, and server-side logic using Node.js and Express.

### **2. LEARNING AND KNOWLEGDE GAAINED**

Today, I learned how to build dynamic web applications by integrating frontend components with backend APIs and performing CRUD operations efficiently.

I also gained practical understanding of Two-Factor Authentication (2FA), OTP verification systems, and the communication flow between frontend and backend applications in a MERN-based architecture.

### **3. PLAN FOR NEXT WORKING DAY**

- Continue working on live projects.
- Explore additional backend development concepts and frontend integrations.

## **DAY 4: 16 April 2026**

### **1. TASKS WORKED ON**

- *Aeko Website – Catalogue Page UI Update:* Updated the button design by removing border radius and aligning the styling with the project's design system to maintain consistency and improve component reusability throughout the application.
- *Netsquare Softwares Website – Theme Customization:* Worked on enhancing the website's visual theme by applying company branding and logo colors across the UI. Updated buttons, icons, images, videos, and other elements with consistent gradient-based styling for a unified design experience.

### **2. LEARNING AND KNOWLEDGE GAINED**

Today, I learned the importance of maintaining design consistency across web applications by applying uniform styling patterns for buttons and UI elements.

I also improved my practical understanding of using gradient-based themes and branding-oriented design approaches to enhance website appearance and visual identity.

### **3. PLAN FOR NEXT WORKING. DAY**

- Continue working on live projects.
- Revise previously learned React concepts and improve implementation practices.

## **DAY 5: 17 April 2026**

### **1. TASKS WORKED ON**

- ***Aeko Website – Catalogue & Terms Page Enhancements:*** Worked on improving the UI and user experience of the Catalogue and Terms & Conditions pages by enhancing styling, readability, spacing, and overall design consistency across the website.
- ***MB-US Trucking Website – Role-Based Control & Security Enhancements:*** Implemented role-based access control and security enhancements by adding confirmation dialogs and restricting sensitive actions based on user roles. Also limited specific editing and management functionalities to Super Admin users only.
- ***Website Testing & Bug Identification:*** Performed testing on the MB Trucking website to verify functionality and application stability. Also identified and documented UI and navigation issues on the Aeko website for further resolution.

### **2. LEARNING AND KNOWLEDGE GAINED**

Today, I improved my understanding of UI/UX design, role-based access control, and permission management in web applications. I also strengthened my testing and debugging skills by identifying and analyzing UI and functional issues across different modules.

### **3. PLAN FOR NEXT WORKING DAY**

- Continue working on ongoing live projects.
- Explore additional UI improvements and backend integration concepts.